

Predictive Modelling and Feature Analysis of Employment Outcomes of International Graduates in the UK

A Statistical and Machine Learning Approach

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Introduction & Research Questions

- International students are a major and growing part of the UK workforce (**685,565** in 2024/25 — HESA)
- Employability after graduation depends on academic, institutional, and demographic factors that are not yet well understood

This project asks:

- 1 How accurately can employment outcomes of international graduates in the UK be predicted using statistical and machine learning classification models?
- 2 Which classification methods provide the best predictive performance for employment outcomes?
- 3 Which factors are the most important predictors of employment outcomes?
- 4 What factors determine salary levels among employed international graduates in the UK?

- UK subset of the international graduate dataset: **74,901** rows
- Research variables: `Employment_Status` (Employed / Unemployed / Continuing Education) and `Salary`
- Predictors include: `GPA`, `Age`, `Years_Since_Graduation`, `Internship_Experience`, `Education_Level`, `University_Ranking`, `Language_Proficiency`, `Field_of_Study`, `Gender`, `Visa_Type`, `Country_of_Origin`

Exploratory Data Analysis

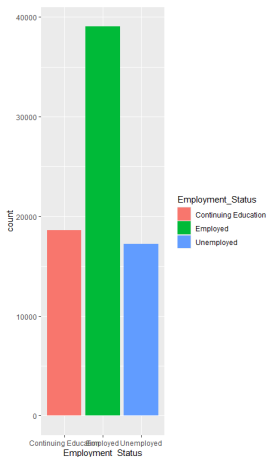


Figure 1: Distribution of Employment Status

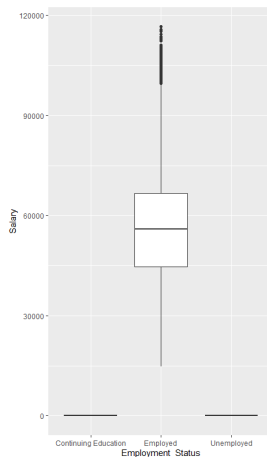


Figure 2: Salary vs Employment Status Distribution

Classification Methods

To answer RQ1 and RQ2, four classifiers were fitted to predict **Employed vs Unemployed** (employment-only predictors used, since Salary alone can dominate):

K-Nearest Neighbours

- Tuned $k = 1, \dots, 50$ via **10-fold cross-validation**, selecting $k = 28$ with highest CV accuracy = 0.8435

Classification Tree, Bagging & Random Forest

- Grew a full tree ($cp = 0$), pruned via the **minimum CV error rule** and the **1-SE rule**; **Bagging** and **Random Forest** fitted as ensembles for comparison

Support Vector Machine

- Linear-kernel SVM

Because the classes are imbalanced, **precision-recall AUC** was used alongside accuracy, rather than relying on ROC-AUC alone.

Classification Results

Table 1: Classifier performance metrics

	KNN	Tree.1SE_rule.	SVM
Accuracy	0.8458189	0.9068693	0.8539945
Sensitivity	0.9448940	0.8676943	0.9157354
Specificity	0.6234141	0.9948097	0.7153979
Precision	0.8492265	0.9973424	0.8783884
F1	0.8945096	0.9280121	0.8966732
PR_AUC	0.9729373	0.9828725	0.9736939
ROC_AUC	0.9377187	0.9548529	0.9376706

Best overall performer: Classification Tree (1SE rule)

Table 2: Confusion Matrix of Classification Tree (1SE-rule)

	Employed	Unemployed
Employed	6755	18
Unemployed	1030	3450

Logistic Regression — Employment Status Model

- Compared **stepwise AIC**, **stepwise BIC**, and **LASSO** (vs. Ridge and Elastic Net) for predictor selection; **LASSO** gave the best test accuracy
- Refit without LASSO's zero-coefficient predictors; **ANOVA** confirmed the reduced LASSO model, BIC, and AIC (Gender removed) converge on the same final predictor set, with near-identical coefficients and p-values
- Adding interaction terms improved fit significantly (ANOVA $p < 2.2 \times 10^{-16}$): **GPA**×**Internship**, **Internship**×**Ranking**, **Education**×**Language**

Table 3: Model comparison

Model	Accuracy
Lasso	0.8538168
Lasso (After variable removal)	0.8533724
AIC (Gender Removed)	0.8533724
BIC	0.8533724

Salary Modelling

- Only employed graduates are used (Salary > 0)
- Two approaches were compared to answer RQ4:
 - **Multiple linear regression** with stepwise BIC selection
 - **Generalised Additive Model (GAM)** allowing a smooth, non-linear effect of GPA

Table 4: Salary model comparison

Model	RMSE	R2
BIC Linear	5457.113	0.9075640
GAM	5300.011	0.9128045

The GAM gives a lower RMSE and higher R^2 , suggesting a non-linear relationship between GPA and salary.

Model Diagnostics

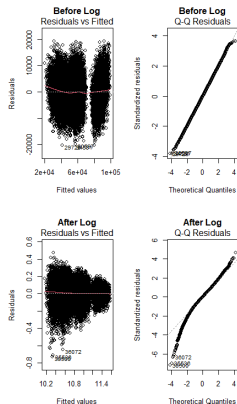


Figure 5: Residual plots for Salary and Log(Salary)

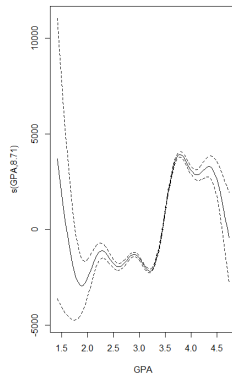


Figure 6: Estimated smooth effect of GPA on salary from the GAM, $s(\text{GPA}, \text{edf} = 8.71)$

- **Classification Tree** achieved the highest predictive performance for employment status (Accuracy = 90.7%, PR-AUC = 0.983).
- **Employment model (logistic)** and **Salary model (GAM)** share the same core predictors: GPA, Internship_Experience, Education_Level, University_Ranking, Language_Proficiency, except Years_Since_Graduation, which matters only for **employment status**
- A **GAM** captures the salary relationship better than a standard linear model, revealing non-linear effects of GPA

Model Insights Dashboard

A quick dashboard highlighting some of the key predictors identified by the logistic regression and GAM models:



Figure 7: Power BI dashboard visualising key model predictors